

An Augmenting-Path Algorithm

correctness follows from
Berge's Thm. and Ex. 13

AUGMENTING-PATH-BIPARTITE($G, \underbrace{(L, R)}$)

a bipartition of G

1. $t \leftarrow 0$
2. $M_t := \emptyset$ $\triangleright$ an empty matching
3. while $\text{NIL} \neq (P_t := \text{FIND-AUGMENTING-PATH-BIP}(G, (L, R), M_t))$
4. $M_{t+1} := M_t \Delta E(P_t)$
5. $t \leftarrow t+1$
6. return M_t

worst-case running time?
 $\text{poly}(n, m)$

FIND-AUGMENTING-PATH-BIP($G, (L, R), M$)

1. $(D(G, M), r, s) \leftarrow \text{BIP-MATCHING-AUXILIARY-DIGRAPH}(G, (L, R), M)$
2. $(\pi, d) := \text{BREADTH-FIRST-SEARCH}(D(G, M), r)$
3. if $d[s] = +\infty$ then return NIL
4. else
5. $\langle v_0, e_1, v_1, \dots, e_\ell, v_\ell \rangle := \text{PATH-FROM-BRANCHING}((\pi, d), s)$
6. return $\langle v_1, \dots, e_{\ell-1}, v_{\ell-1} \rangle$

The Matching Number of a Graph

Let $G := (V, E, \varphi)$ be a graph.

read as "nu of G"

The **matching number** of G , denoted by $\nu(G)$, is the largest size of a matching in G .

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⋖ ← is it possible to convince us of this?

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Q: is there an object, that may be returned along with M ,

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How does one get convinced of NON-existence
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Matching vs. Vertex Covers

Proposition 14 Let $G := (V, E, \varphi)$ be a graph.

Let $M \subseteq E$ be a matching and let $K \subseteq V$ be a vertex cover of G .

Then $|M| \leq |K|$.

Matching vs. Vertex Covers

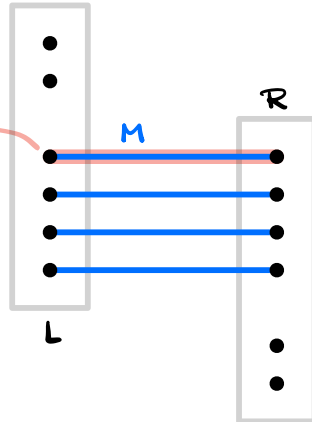
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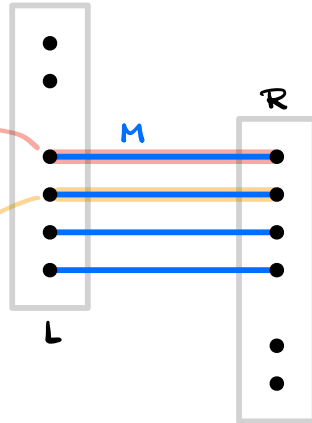
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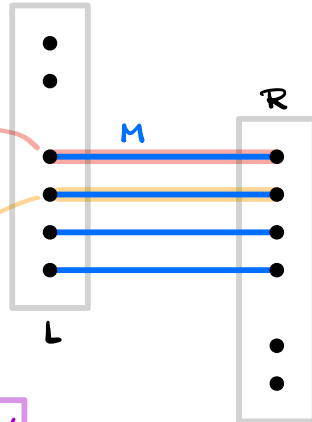
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because these edges are disjoint



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Proof There is a function $f: M \rightarrow K$ that sends each edge $e \in M$ to one of its ends in $\varphi(e) \cap K \neq \emptyset$. Since $\{\varphi(e) \mid e \in M\}$ is pairwise disjoint, f is injective. □

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Matching vs. Vertex Covers

Theorem 16 Let X be a nonempty finite set and let $f: X \rightarrow \mathbb{R}$ be a function. Then there exist $x_* \in X$ and $x^* \in X$ s.t., for each $x \in X$, one has $f(x_*) \leq f(x) \leq f(x^*)$.

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BE CAREFUL: this is often overlooked

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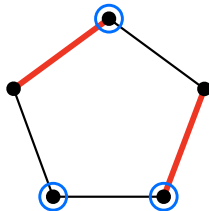
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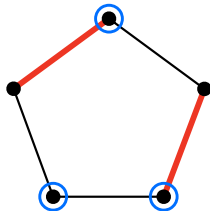
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Exercise 17 Let $G := (V, E, \varphi)$ be a graph and let $M \subseteq E$ be an inclusion-wise maximal matching. Prove that V_M is a vertex cover of G , and conclude that $\tau(G) \leq 2\nu(G)$.

Matching Number Upper Bound

maybe a vertex cover K of G
of size $|K| = |M|$

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EVEN YOUR BOSS
would understand this

Verification Procedure

VERIFY-MATCHING-AND-VERTEX-COVER (G, M, K)

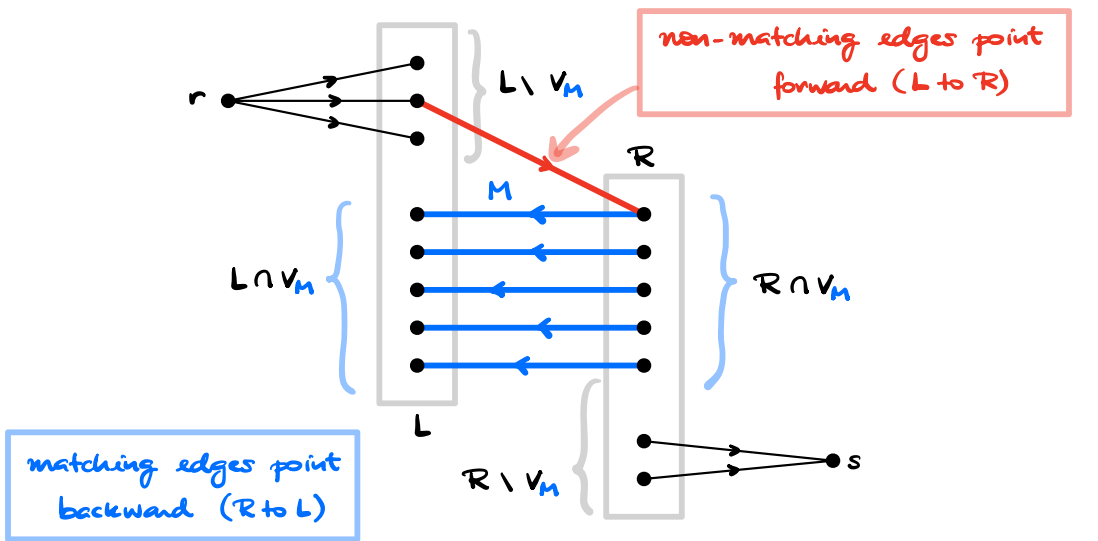
1. VERIFY-MATCHING (G, M)
2. VERIFY-VERTEX-COVER (G, K)
3. return $|M| = |K|$

Verification Procedure - Alternatively

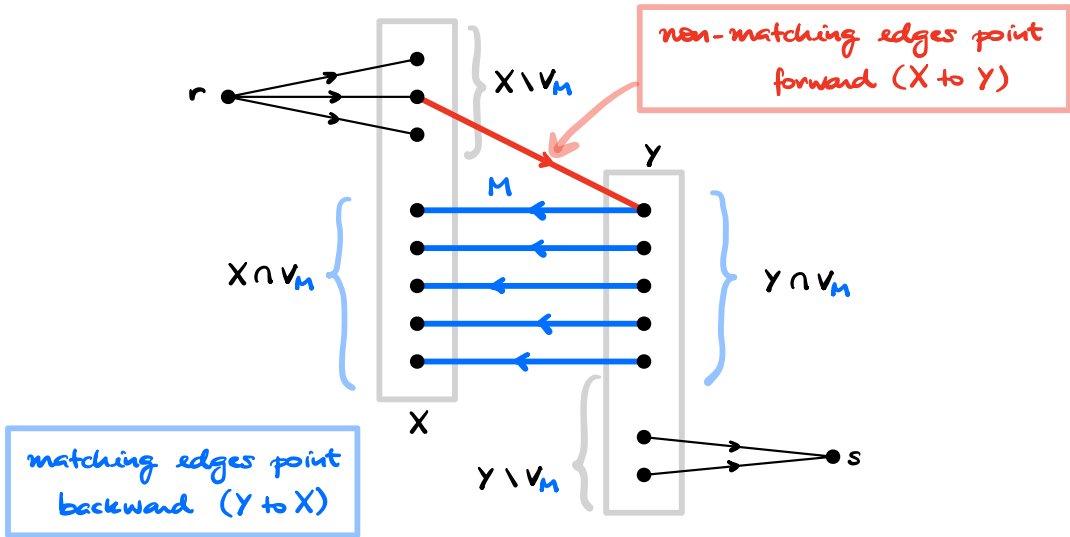
VERIFY-MATCHING-AND-VERTEX-COVER (G, M, K)

1. $V_M \leftarrow \emptyset$ $\triangleright V_M \subseteq V(G)$
2. for each $e \in E(G)$ do
3. $\{u, v\} \leftarrow e$
4. if $e \in M$ then
5. if $u \in V_M$ or $v \in V_M$ then return FALSE
6. $V_M \leftarrow V_M \cup \{u, v\}$
7. if $u \notin K$ and $v \notin K$ then return FALSE
8. return $|M| = |K|$

Vertex Covers from Reachables



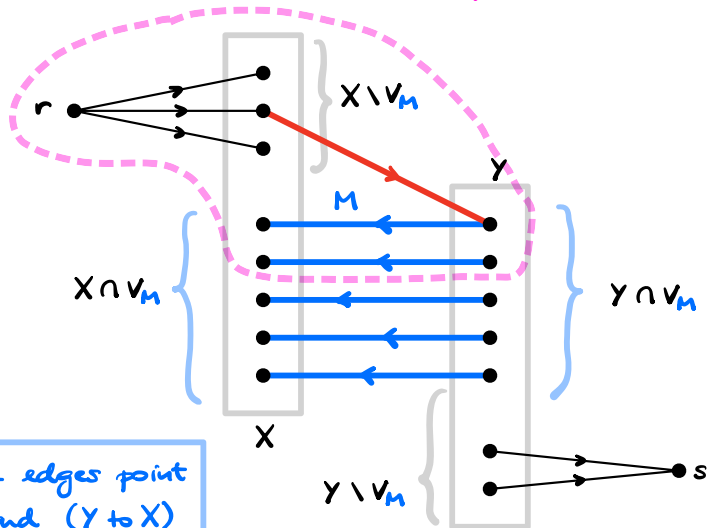
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$$R_D := \{v \in V(D) : r \rightsquigarrow_D v\}$$



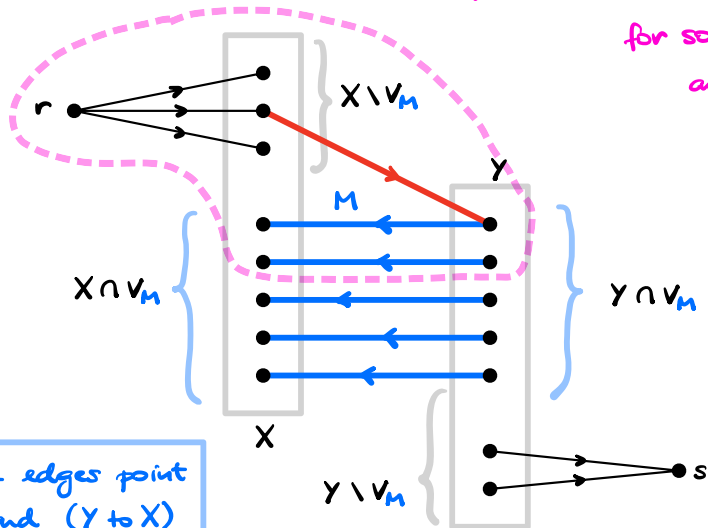
matching edges point
backward (Y to X)

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$\mathcal{D} := \mathcal{D}(G, M)$

$$R_D := \{v \in V(D) : r \rightsquigarrow_D v\} = \{r\} \cup R \cup S$$

for some $R \subseteq V(G)$
and $S \in \{\emptyset, \{s\}\}$



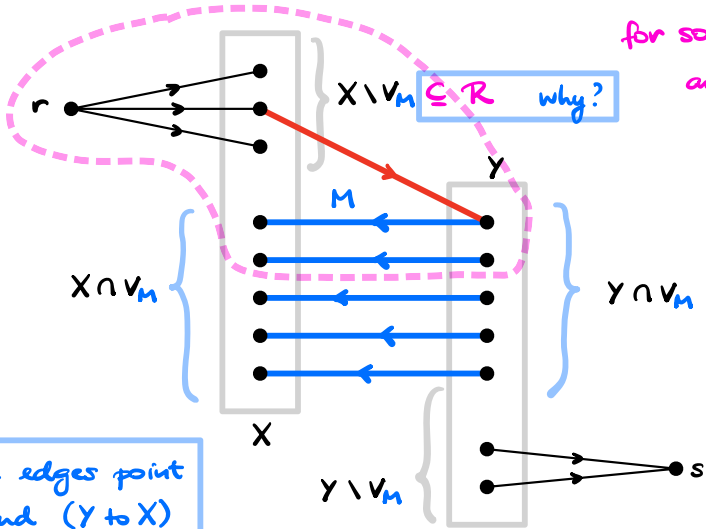
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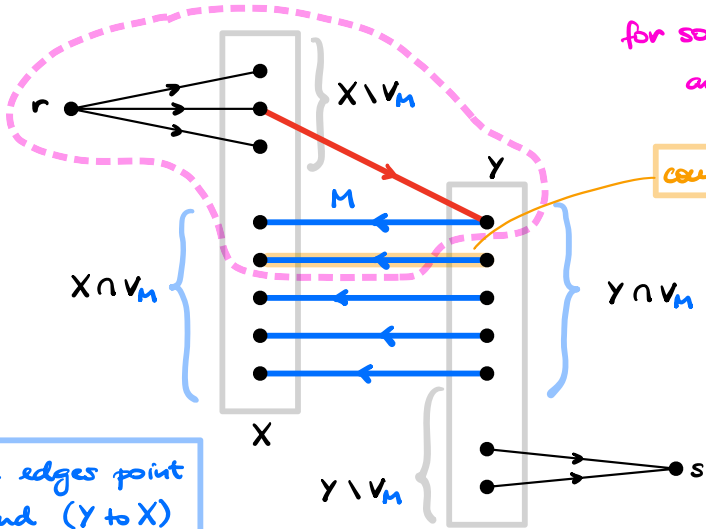
$X \setminus V_M \subseteq R$ why?

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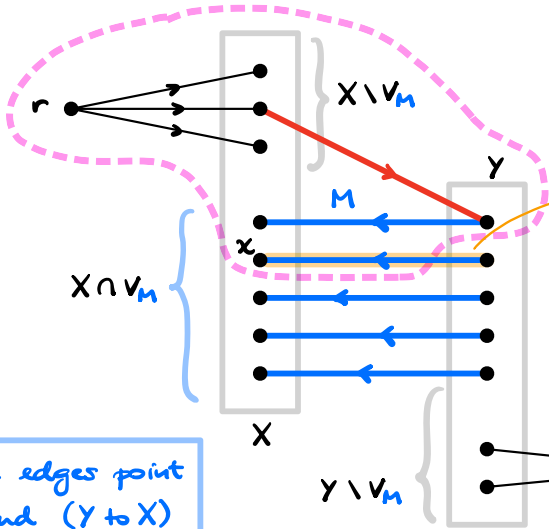
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could this happen?

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could this happen?

NO, this is the only
arc entering x

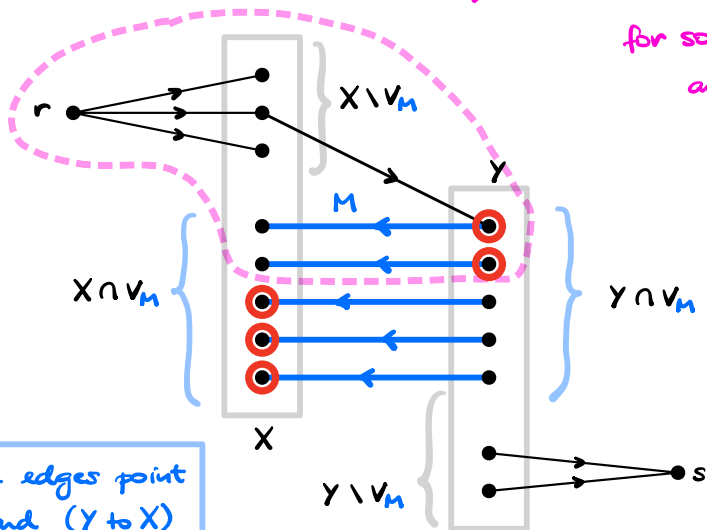
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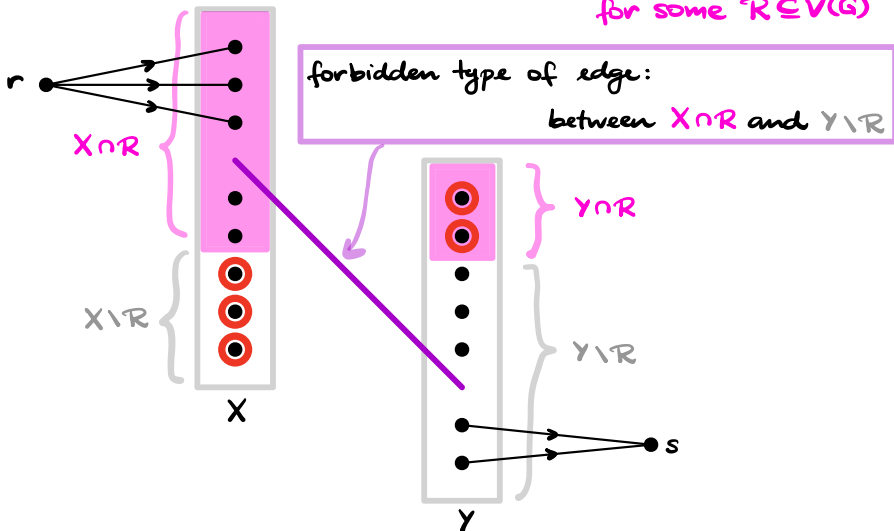
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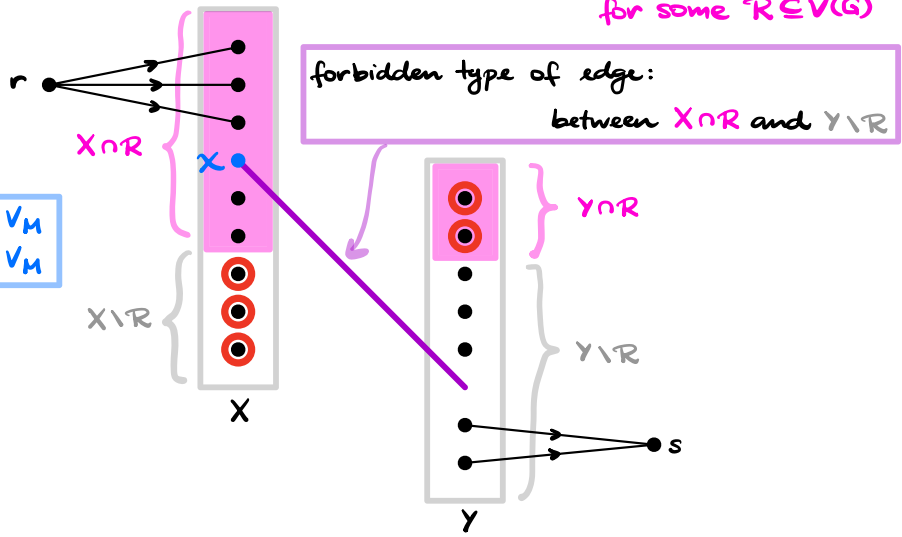
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- Case 2: $x \in V_M$



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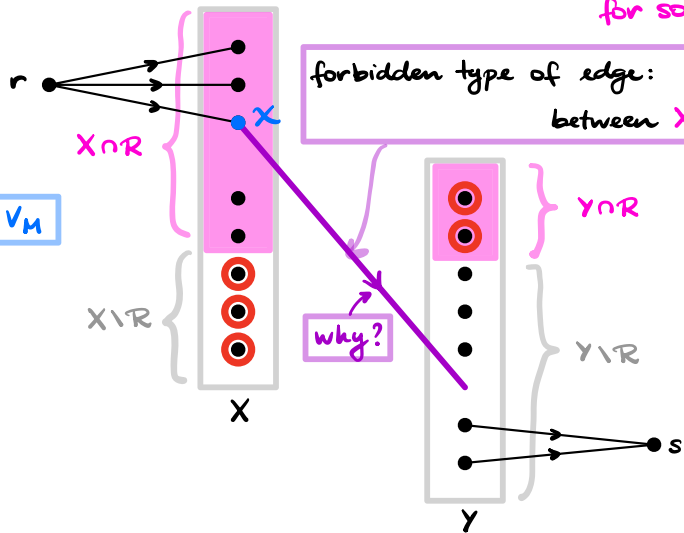
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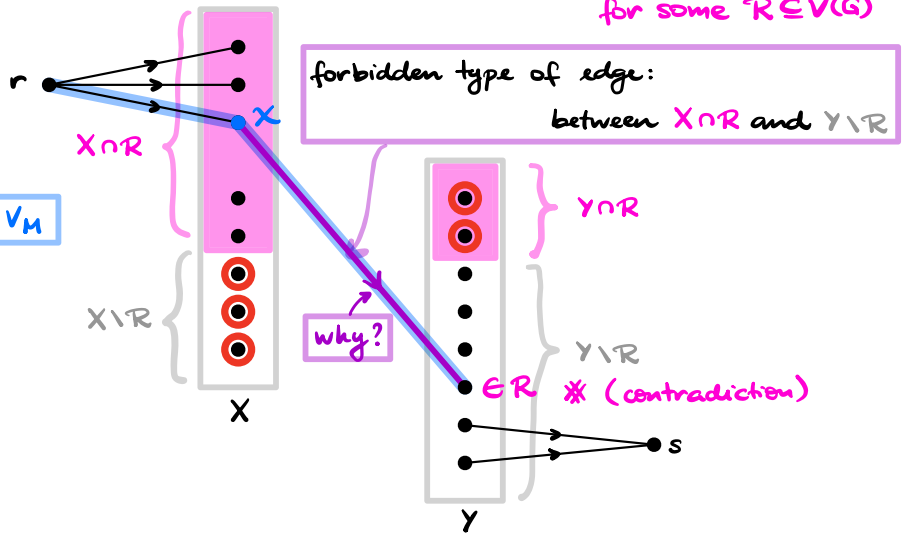
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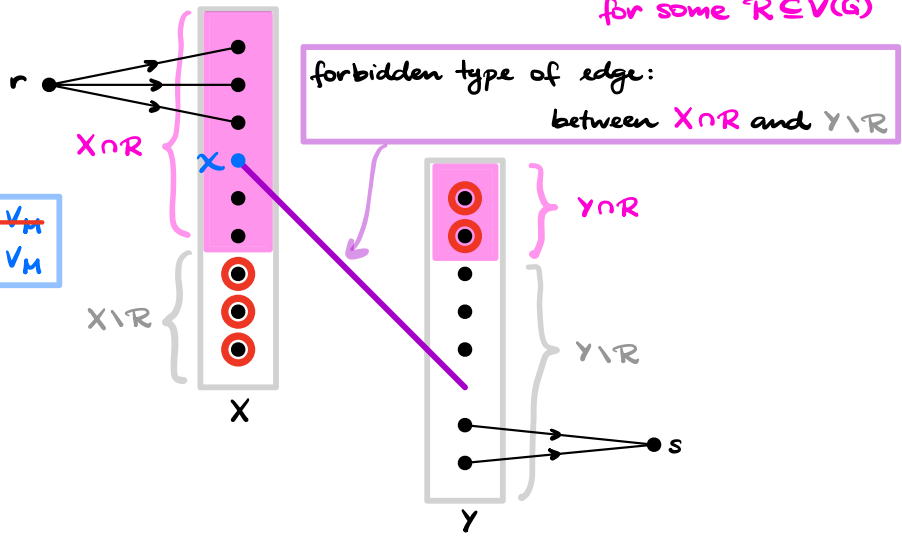
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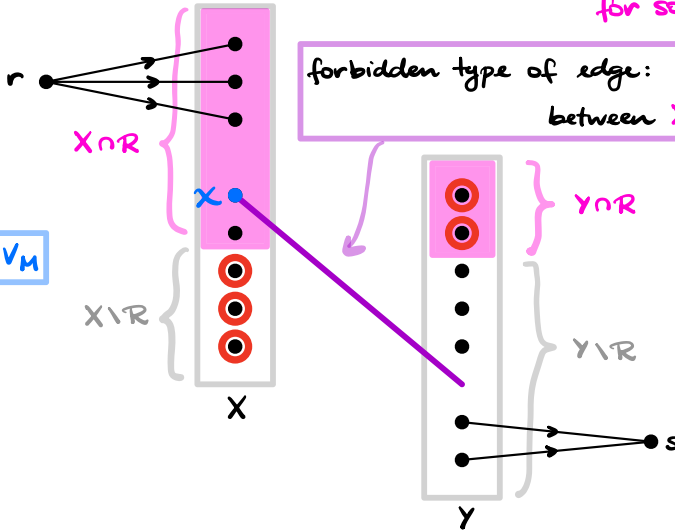
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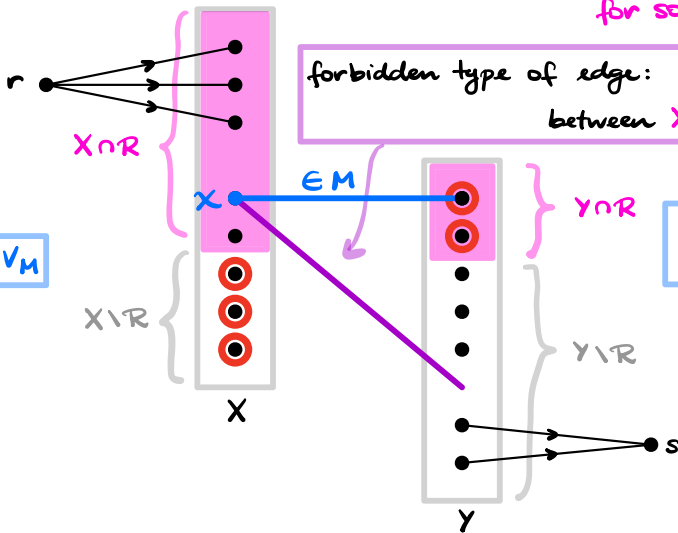
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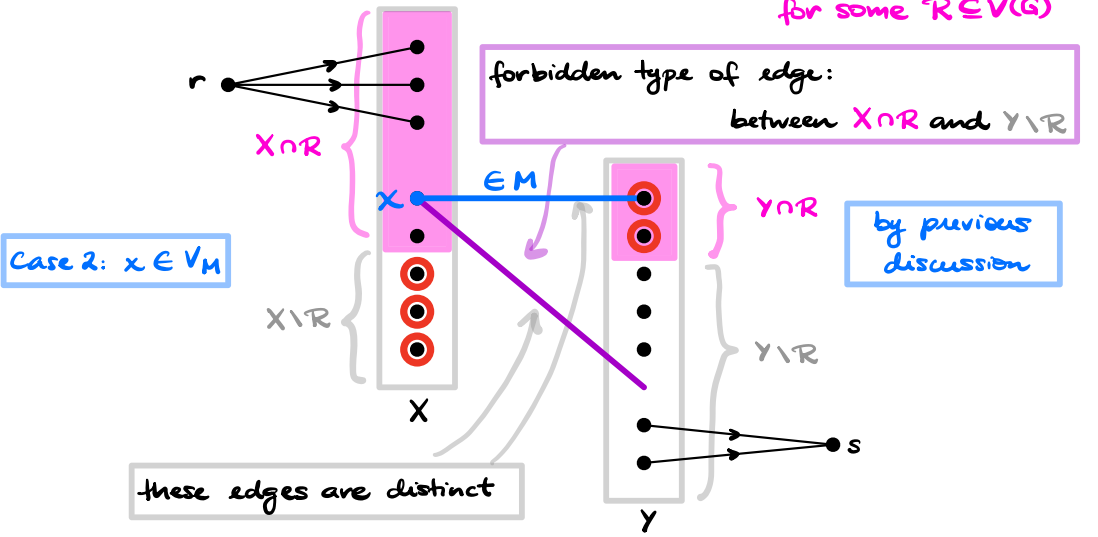
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Vertex Covers from Reachables

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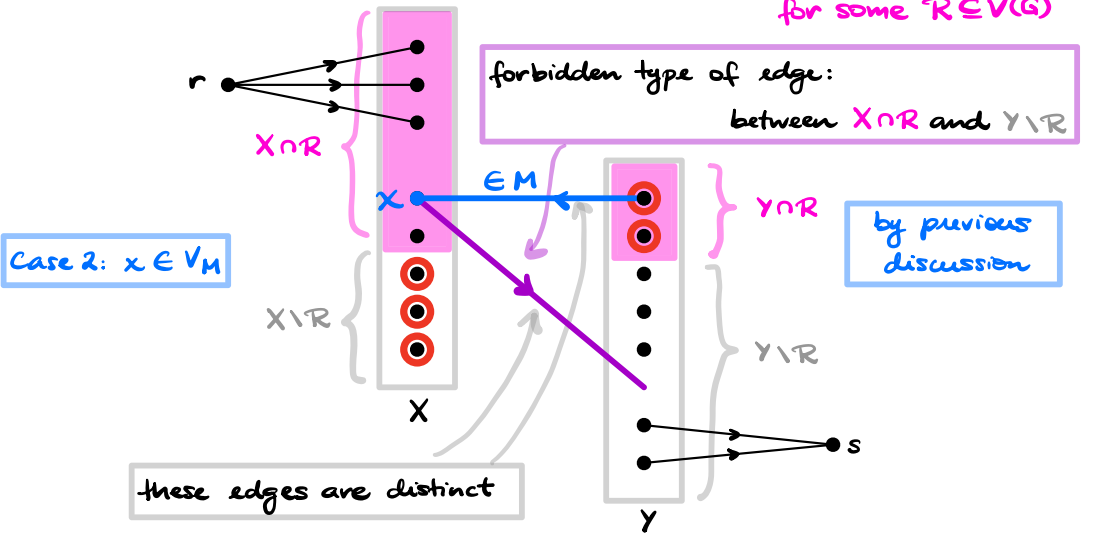
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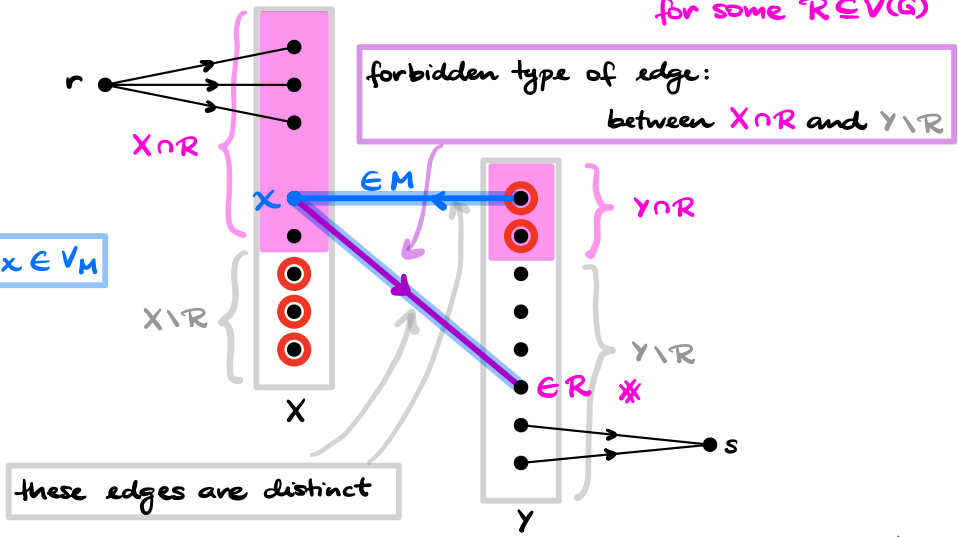
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Case 2: $x \in V_M$



forbidden type of edge:
between $X \cap R$ and $Y \setminus R$

proved!

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Exercise 18 Let $G := (V, E, \varphi)$ be an (X, Y) -bipartite graph. Let $M \subseteq E$ be a matching. Set $R_D := \{v \in D(G, M) : r \rightsquigarrow_{D(G, M)} v\}$, so there is $R \subseteq V$ s.t. $R_D = \{r\} \cup R$. Prove that $K := (X \setminus R) \cup (Y \cap R)$ is a vertex cover.

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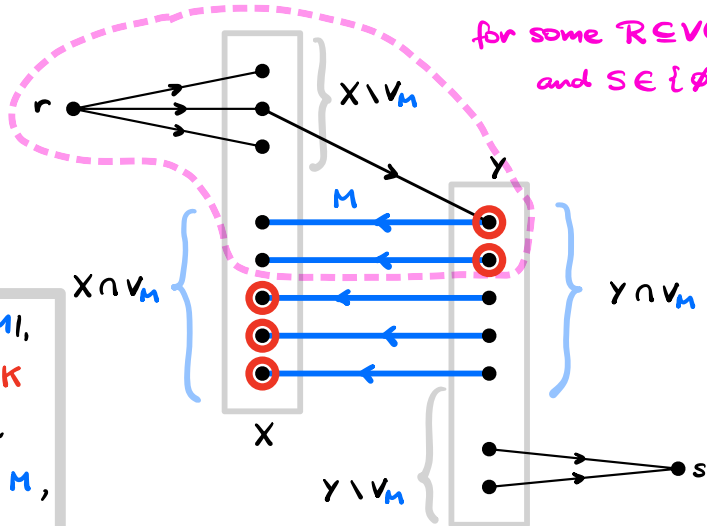
Comment: lecture proofs vs written proofs (for grading)

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if we want $|K| = |M|$,
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precisely one end
of each edge $e \in M$,
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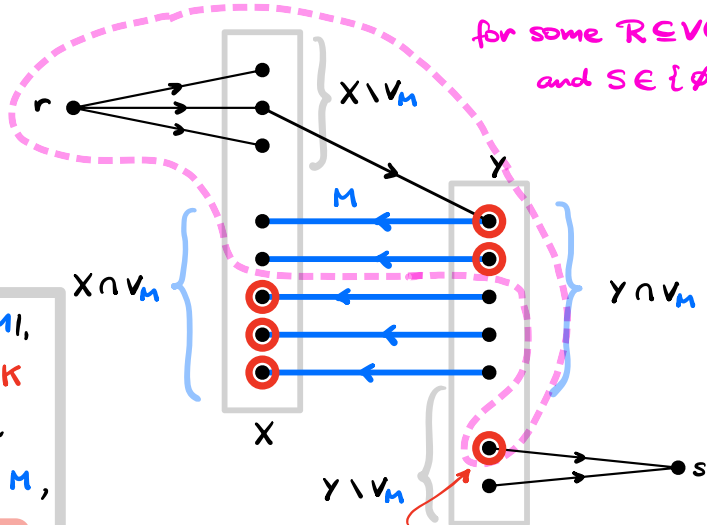
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Vertex Covers from Reachables

$$\mathcal{D} := \mathcal{D}(G, M)$$

$$\mathcal{R}_{\mathcal{D}} := \{v \in V(\mathcal{D}) : r \rightsquigarrow_{\mathcal{D}} v\} = \{r\} \cup R \cup S$$

for some $R \subseteq V(G)$
and $S \in \{\emptyset, \{s\}\}$



if we want $|K| = |M|$,
we must add to K
precisely one end
of each edge $e \in M$,
and nothing else

if r reaches s , then $|K| > |M|$

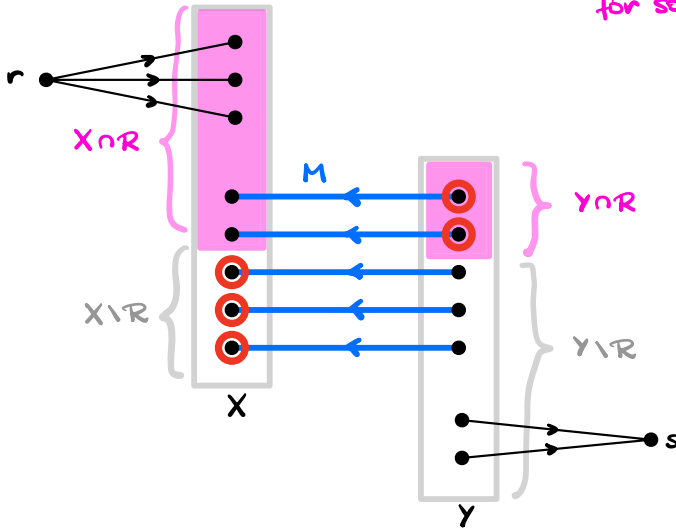
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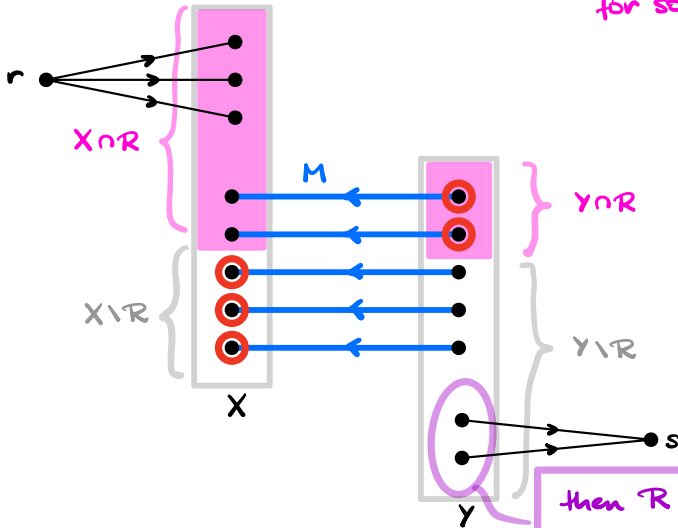
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then R and $Y \setminus V_M$ are disjoint

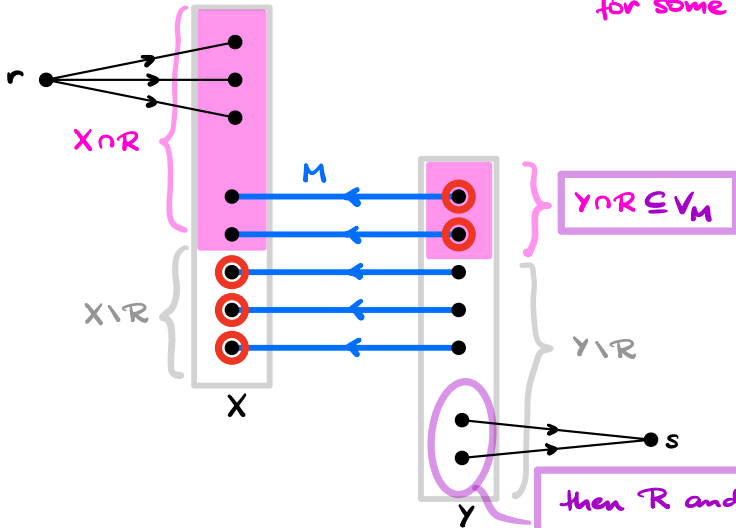
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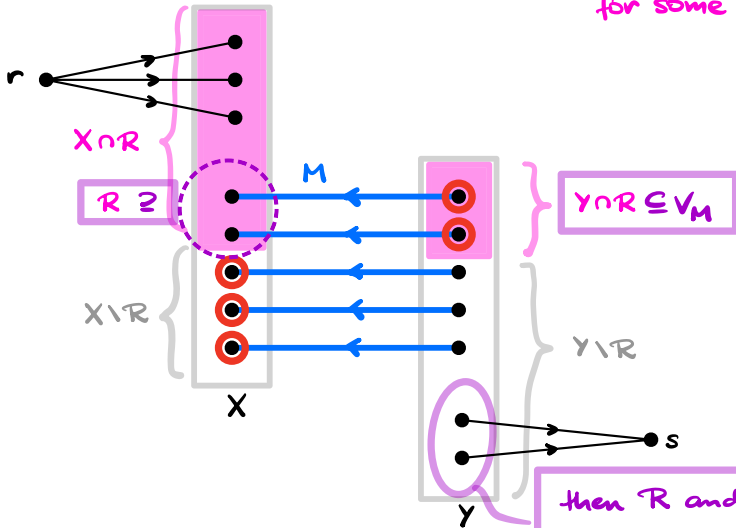
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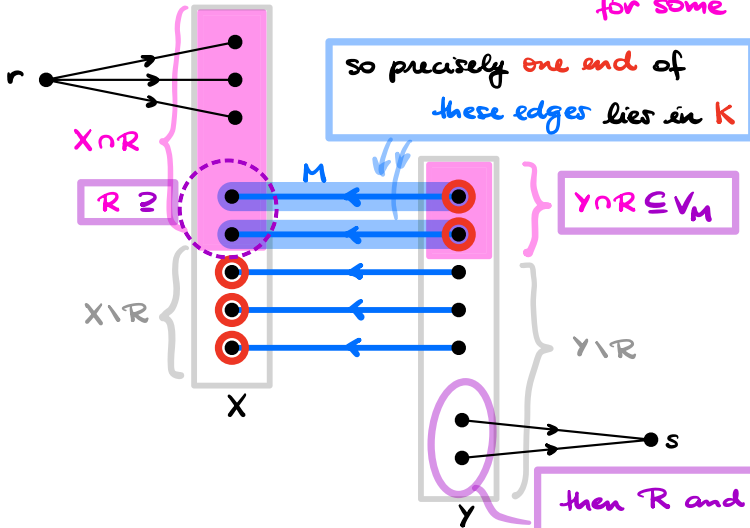
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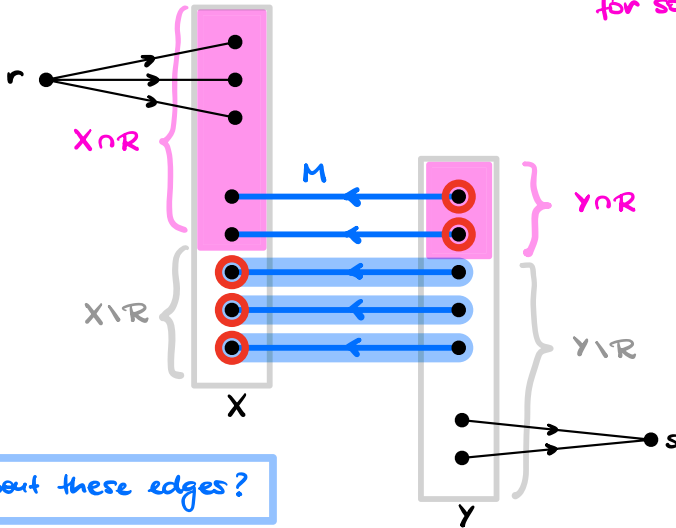
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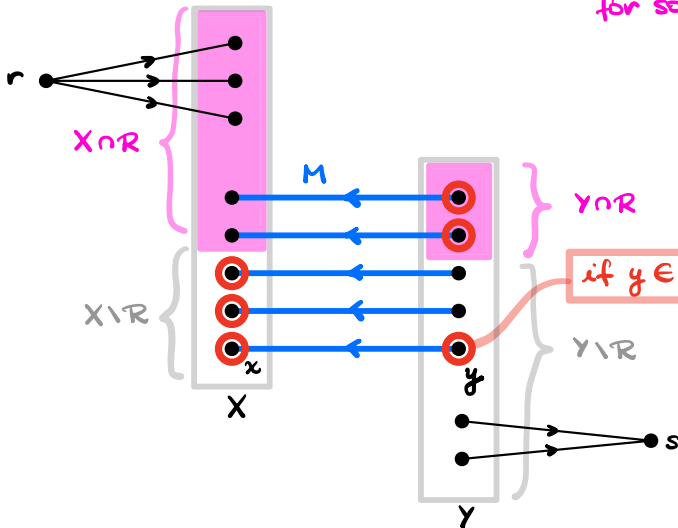
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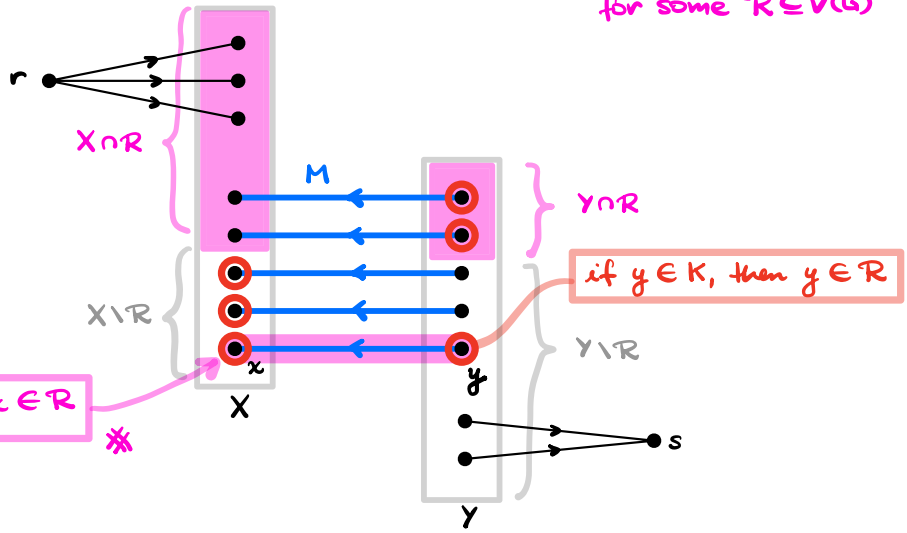


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Vertex Covers from Reachables

$D := D(G, M)$

Assume $s \notin R_D := \{v \in V(D) : r \rightsquigarrow_D v\} = \{r\} \cup R$
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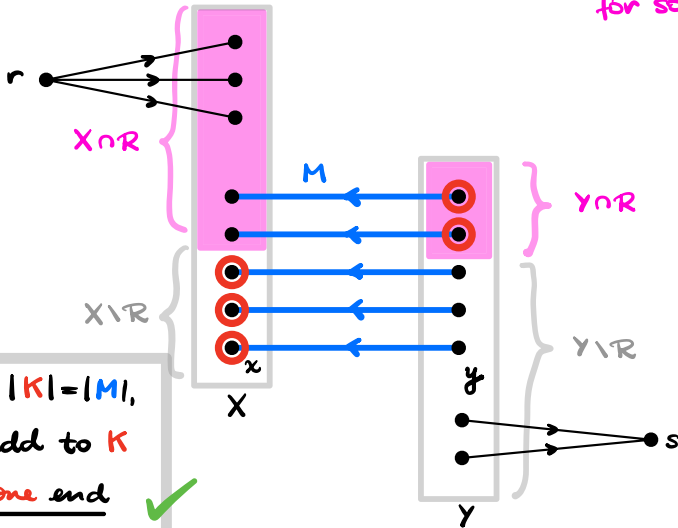
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Vertex Covers from Reachables

Theorem 19 Let $G := (V, E, \varphi)$ be an (X, Y) -bipartite graph. Let $M \subseteq E$ be a matching. Set $R_D := \{v \in D(G, M) : r \rightsquigarrow_{D(G, M)} v\}$, so there is $R \subseteq V$ s.t. $R_D \setminus \{s\} = \{r\} \cup R$. Set $K := (X \setminus R) \cup (Y \cap R)$. If $s \notin R_D$, then $|K| = |M|$.

Proof From discussion above. □

König's Algorithm

KÖNIG'S-ALGORITHM($G, \underbrace{(X, Y)}$)

1. $t \leftarrow 0$

a bipartition of G

2. $M_t := \emptyset$ $\triangleright$ an empty matching

3. while 42 do

4. $(D_t, r, s) := \text{BIP-MATCHING-AUXILIARY-DIGRAPH}(G, (X, Y), M_t)$

5. $T_t := \text{BREADTH-FIRST-SEARCH}(D_t, r)$; $R_t := \text{REACHABLE-SET}(T_t)$

6. if $s \notin R_t$ then return $(M_t, (X \setminus R_t) \cup (Y \cap R_t))$

7. $\langle v_0, e_1, v_1, \dots, e_\ell, v_\ell \rangle := \text{PATH-FROM-BRANCHING}(T_t, s)$

8. $P_t := \langle v_1, \dots, e_{\ell-1}, v_{\ell-1} \rangle$

9. $M_{t+1} := M_t \Delta E(P_t)$

10. $t \leftarrow t+1$

REACHABLE-SET(T)

1. $(_, d) := T$

2. return $\{v \in V : d[v] < +\infty\}$

König's Algorithm

Corollary 20 Let $G := (V, E, \varphi)$ be an (X, Y) -bipartite graph.

Then the call **KÖNIG'S-ALGORITHM** $(G, (X, Y)) =: (M, K)$ terminates after $\leq |V|$ iterations of the while loop, M is a maximum matching of G , the set K is a minimum vertex cover of G , and $|M| = |K|$.

Proof Each M_t is a matching by induction on t and **Ex. 13**. The iteration bound follows since $|M_t|$ is a strictly increasing natural number bounded above by $|V|$. The set K is a vertex cover of G by **Ex. 18** and $|M| = |K|$ by **Thm. 19**. For each matching $N \subseteq E$ of G , one has $|N| \leq |K| = |M|$, so M is a maximum matching. For each vertex cover L of G , one has $|L| \geq |M| = |K|$, so K is a minimum vertex cover. □

König's Matching Theorem

Corollary 21 (König's Matching Theorem) Let G be a bipartite graph. Then $\tau(G) = \nu(G)$, i.e., the minimum size of a vertex cover is equal to the maximum size of a matching.

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Examples of Minmax Relations

Max-Potential Min-Work Theorem Let $\mathcal{D} := (V, A, \varphi)$ be a digraph, and let $r, s \in V$. Then the minimum length of an rs -path is equal to the maximum number of pairwise disjoint rs -cuts.

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these are not numbered because you cannot (yet) use them

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Menger's Theorem Let $\mathcal{D} := (V, A, \varphi)$ be a digraph, and let $r, s \in V$. Then the minimum size of an rs -cut is equal to the maximum number of pairwise arc-disjoint rs -paths.

Examples of Minmax Relations

Mirsky's Theorem Let $(P, \preceq)$ be a poset. Then the minimum number of antichains covering P is equal to the maximum size of a chain.

Dilworth's Decomposition Theorem Let $(P, \preceq)$ be a poset. Then the minimum number of chains covering P is equal to the maximum size of an antichain.

Examples of Minmax Relations

König's Matching Theorem Let G be a bipartite graph. Then the minimum size of a vertex cover is equal to the maximum size of a matching.

König-Rado Edge Cover Theorem Let G be a bipartite graph without isolated vertices. Then the minimum size of an edge cover is equal to the maximum size of a stable set.

Examples of Minmax Relations

König's Edge Coloring Theorem Let G be a bipartite graph. Then the minimum number of matchings covering $E(G)$ is equal to the maximum degree of a vertex.

Gupta's Theorem Let G be a bipartite graph. Then the minimum degree of a vertex is equal to the maximum number of pairwise disjoint edge covers.

Examples of Minmax Relations

Tutte-Berge Formula Let $G := (V, E, \varphi)$ be a graph. Then the maximum size of a matching is equal to

$$\min_{S \subseteq V} \frac{1}{2} (|V| + |S| - \text{odd}(G-S)).$$

$\text{odd}(\cdot) :=$ number of odd components of $\cdot$

Theorem Let $G := (V, E, \varphi)$ be a graph without isolated vertices.

Then the minimum size of an edge cover is equal to

$$\max_{S \subseteq V} \frac{1}{2} (|S| + \text{odd}(G[S])).$$

Examples of Minmax Relations

Tutte - Nash-Williams Disjoint Trees Theorem Let $G := (V, E, \varphi)$ be a

graph. Then the maximum number of pairwise-disjoint spanning

trees of G is $\min_{\mathcal{P}} \left\lfloor \frac{|\delta(\mathcal{P})|}{|\mathcal{P}| - 1} \right\rfloor$, $\delta(\mathcal{P}) := \{e \in E : \forall P \in \mathcal{P}, \varphi(e) \notin P\}$

where $\mathcal{P}$ ranges over all partitions of V st. $|\mathcal{P}| \geq 2$.

Nash-Williams' Covering Forests Theorem Let $G := (V, E, \varphi)$ be a graph.

Then the minimum number of forests covering E is

$$\max_{\emptyset \neq S \subseteq V} \left\lceil \frac{|E_{\varphi}[S]|}{|S| - 1} \right\rceil$$

Examples of Minmax Relations

i.e., its underlying graph is connected

The Lucchesi-Younger Theorem Let $D := (V, A, \varphi)$ be a weakly connected digraph. Then the **minimum size of a dijoin** is equal to the **maximum number of pairwise disjoint directed cuts**.

a set of arcs that intersects every directed cut

Woodall's Conjecture Let $D := (V, A, \varphi)$ be a digraph. Then the **minimum size of a directed cut** is equal to the **maximum number of pairwise disjoint dijoins**.